

KAVEESHA DHANANJAYA

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I am a dedicated Mechatronics Engineering student with a strong academic record and hands-on experience in robotics and automation. Skilled in mechanical design, electrical modeling, and programming, with expertise in designing and fabricating robotic systems for real-world applications. Proficient in advanced modeling tools like SolidWorks and MATLAB, with hands-on experience in programming and integrating PLCs for industrial automation. Recognized for leadership roles in academic and extracurricular settings, I bring a collaborative mindset and a passion for innovative technology solutions. Eager to contribute to dynamic engineering projects that drive automation and intelligent design.

EDUCATION

- BSc Eng (Hons) in Mechatronics Engineering,** **08/2021 – Present**
General Sir John Kotelawala Defense University, Ratmalana
- Cumulative GPA: 3.91 / 4.0
 - Key modules: Advanced Robotics, Machine Elements in Design, Power Electronics, Digital Signal Processing
- GCE Advanced Level,** Royal College, Colombo **2020**
- Z- Score : 1.4

WORK EXPERIENCE

- Senior Robotics Engineer,** Hype Insight (Pty) Ltd **9/2025 – present**
- Design and development of four wheel robot integrating AI for computer vision.
 - Research and development of Rover and drone concepts .
 - Building Mic array by developing an audio processing pipe-line.
- Product Development Engineer - Innovations,** MAS Capital (Pvt) Ltd, Biyagama **4/2025 – 8/2025**
- Fabrication and Development of Mechanical systems to enhance process Efficiency.
 - Research and Innovation on new technologies to Develop Products .
 - Feasibility Analysis to develop new Concepts.
- Tutor - 360 Labs,** Borelasgamuwa (Hybrid) **12/2023 – 12/2024**
- Designed STEM kits and led workshops to engage students aged 5–20 in STEM concepts.
 - Delivered robotics courses in schools to spark interest in technology.
 - Completed automation projects to advance agricultural practices.
- Intern (Autonomation Engineer) - Autonomation Lab,** MAS Capital (Pvt) Ltd, Ratmalana **10/2023 – 03/2024**
- Assisted in the design, fabrication, testing, validation, and documentation processes of new developments.
 - Designed and fabricated three modular units using PLC for strap making, ironing and cutting.
 - Developed a Convolution Neural Network to Identify the presence and orientation of a bra cup.

PROJECT EXPERIENCE

- Quadruped Robot for Inspection in Human-Inaccessible Structured Areas,** KDU, Ratmalana **Ongoing**
- Four-legged robot equipped with a five-bar mechanism for each leg to navigate structured terrains.
 - Responsibilities: MATLAB simulation, ROS2 communication, wiring diagrams, inverse kinematic modeling
- Mars Rover-Inspired Wheeled Robot with LiDAR for Mapping and Navigation,** KDU, Ratmalana **2023**
- Designed and fabricated a Mars Rover-like wheeled robot equipped with LiDAR which utilizes SLAM algorithm on ROS to map, localize, and navigate in a structured environment.
 - Responsibilities: Kinematics modeling, GAZEBO simulation, 3D Modeling, Electronics Assembly
- Sweep the Table Robot Contest,** KDU, Ratmalana **2023**
- Designed a wheeled robot to autonomously push objects off a rectangular arena using a rotating hammer.
 - 1st place and "Best Design" award at KDU FOE Intake 38 Mechatronics Systems Design Competition.
 - Responsibilities: Developing pushing mechanisms Electronics Assembly, Fabrication, Programming

PUBLICATIONS

Design & Development of a Quadruped Robot for Inspection in Human Inaccessible Structured Areas Publication Under Progress	2025
Prediction of Coronary Artery Disease using Artificial Neural Network KDU- 17th International Research Conference	2024
Development of an Automated Clothesline System KDU- 16th International Research Conference	2023

EXTRACURRICULAR ACTIVITIES

Research In-charge , Electronic Robotics and Innovation Club, KDU, Ratmalana	12/2022 – 12/2023
Senior Prefect , Royal College, Colombo	12/2020 – 12/2021
President Scout , Royal College, Colombo	12/2022
Captain, Football , Royal College, Colombo	2014, 2016, 2017

CERTIFICATIONS AND TRAININGS

Machine Learning , DeepLearning.AI, Stanford University - Coursera	07/2023 – 08/2023
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SKILLS AND LANGUAGES

PLC & HMI Programming: ISPSoft, SIMATIC STEP7, SKTool, SATool, Nextion IDE
Mechanical Modeling and Designing: SolidWorks, Fusion360, Ansys, MATLAB
Microcontrollers and processors: SolidWorks, Fusion360, Ansys, MATLAB
Simulation Software: MATLAB (Simulink, Simscape) , GAZEBO(ROS2 Environment)
Electrical Modeling and Designing: Proteus, EasyEDA, KiCAD, LTSpice, Eagle
Programming Languages: Python, C++
Languages: English (Fluent), Sinhala (Native)

HOBBIES AND INTERESTS

RC Car Racing, Football, 3D Printing

REFERENCE

Mr. Isira Naotunna

BSc. Hons(Moratuwa, Sri Lanka), M.Eng(CMU, Thailand)

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